

Chapter 84

Using Open FX, Resolve FX, and Fuse Plugins

Fusion’s capabilities can be extended using different kinds of plugins. All compositions in Fusion Studio and in the Fusion page of DaVinci Resolve support third-party Open FX plugins.

Additionally, the Fusion page of DaVinci Resolve provides access to all of the Resolve FX that come with DaVinci Resolve.

Lastly, you can develop your own plugins without using a computer development environment by scripting Fusion’s native Fuse plugins.

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What Are Open FX?

Fusion is able to use compatible Open FX (OFX) plugins that are installed on your computer. Open FX is an open standard for visual effects plugins. It allows plugins written to the standard to work on both DaVinci Resolve and Fusion Studio as well as other applications that support the standard.

OFX plugins can be purchased and downloaded from third-party suppliers such as BorisFX, Red Giant, and RE:Vision Effects. All OFX appear in the Open FX category of the Effects Library, alongside all other effects that are available in Fusion.

What Are Resolve FX?

The Fusion page in DaVinci Resolve is also able to access Resolve FX. Resolve FX are filter effects within DaVinci Resolve. Although most Resolve FX have the same capabilities on all DaVinci Resolve pages, some are specific to the Color page and require the use of the Color page tracker for full functionality. All Resolve FX appear in the Fusion page Effects Library in the Open FX category alongside third-party OFX plugins. Resolve FX are not available in Fusion Studio.

For more information about Resolve FX, see *Chapter 154, "Resolve FX."*

Applying Open FX and Resolve FX Plugins

Resolve FX and OFX plugins are applied to the Node Editor just as you would apply native Fusion nodes. They can help you create fast, interesting looks and effects using the Color page, or imaginative transitions and effects on your clips on the Edit page. Resolve FX are installed with DaVinci Resolve,

After installing a set of OFX plugins, you can access them or Resolve FX plugins in Fusion by opening the Open FX category in the Effects Library.

To add a plugin to the Node Editor, either click the Open FX or Resolve FX plugin name in the Effects Library or drag and drop the plugin onto a connection line to insert it into the node tree. If the plugin has editable settings, you can adjust these in the Inspector.

DaVinci Resolve OFX Renderer for Fusion Studio

You can use the OFX DaVinci Resolve Renderer plugin in Fusion Studio. This plugin allows Fusion Studio to open and apply DaVinci Resolve's color changes using exported DRX stills. Essentially this turns an installation of DaVinci Resolve Studio into one large OFX plugin. This can be very useful when round tripping between DaVinci Resolve Studio and Fusion Studio, where you may want to maintain the exact look created in DaVinci Resolve.

Exporting a DRX still contains much more information about the grade than a simple LUT. DRX files can include data like all the native color and sizing palettes and other items you stored in a Gallery still. This ensures that the grade the colorist created can be exactly replicated in the VFX software.

As of this writing, the following attributes WILL transfer via the DaVinci Resolve OFX Renderer Plugin:

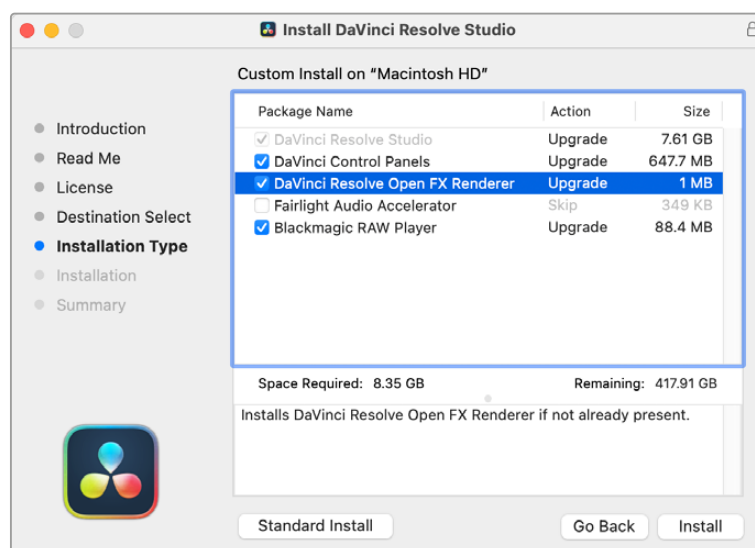
- All native color palettes (Primaries, HDR, Curves, etc.).
- All native sizing palettes (Input, Output, Node, etc.).
- Most Resolve FX.
- Most third-party Open FX.

As of this writing, the following attributes WILL NOT transfer via the DaVinci Resolve OFX Renderer Plugin:

- Resolve FX and Open FX that are temporal based.
- Magic Mask.

To set up the DaVinci Resolve OFX Renderer Plugin:

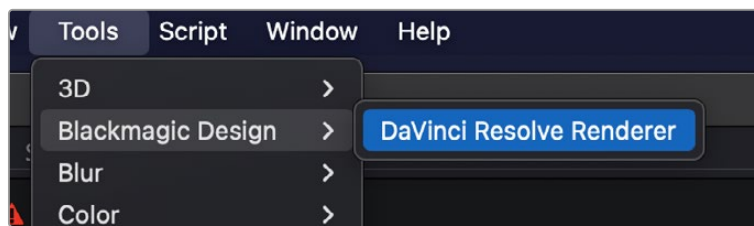
- 1 When installing DaVinci Resolve Studio, select Custom Install and then check the plugin option in the program installer.



Select Custom Install, then check DaVinci Resolve Open FX Renderer to install the plugin

- 2 Create your grade for a clip in the Color page, then save a still to the Gallery. From the Gallery, export a .DRX file.

- 3 In Fusion Studio, apply the DaVinci Resolve Renderer plugin where appropriate for your composition.

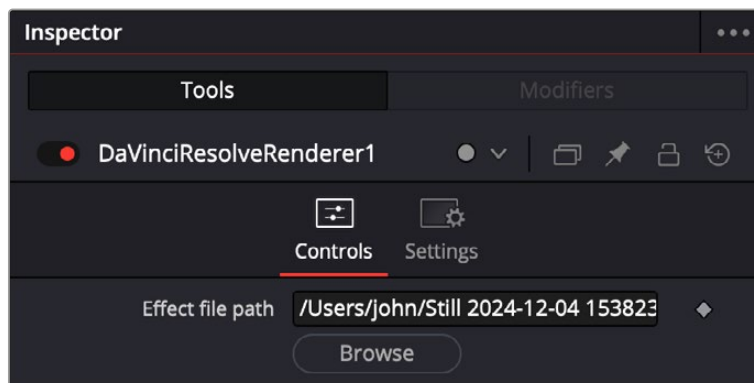


ApplyRendererPluginFusion



Inserting the DaVinci Resolve Renderer OFX into a Fusion Studio node tree

- 4 In the DaVinci Resolve Renderer Inspector in Fusion Studio, choose the .DRX file you exported from the file browser.



Select the .DRX file you exported from DaVinci Resolve in the plugin

Introduction to Fuse Plugins

Fuses are plugins developed for Fusion using the Lua built-in scripting language. Being script-based, Fuses are compiled on-the-fly in Fusion without the need of a computer programming environment. While a Fuse may be slower than an identical Open FX plugin created using Fusion's C++ SDK, a Fuse will still take advantage of Fusion's existing nodes and GPU acceleration.

To install a Fuse:

- 1 Use the .fuse extension at the end of the document name.
- 2 For DaVinci Resolve, save it in one of the following locations:

On macOS: Macintosh HD/Users/username/Library/Application Support/Blackmagic Design/DaVinci Resolve/Fusion/Fuses

On Windows: C:\Users\username\AppData\Roaming\Blackmagic Design\DaVinci Resolve\Support\Fusion\Fuses

On Linux: home/username/.local/share/DaVinciResolve/Fusion/Fuses

For Fusion Studio, save it in one of the following locations:

On macOS: Macintosh HD/Users/username/Library/Application Support/Blackmagic Design/Fusion/Fuses/

On Windows: C:\Users\username\AppData\Roaming\Blackmagic Design\Fusion\Fuses

On Linux: home/username/.fusion/BlackmagicDesign/Fusion/Fuses

You can open and edit Fuses by selecting the Fuse node in the Node Editor and clicking the Edit button at the top of the Inspector. The Fuse opens in the text editor specified in the Global Preferences/Scripting panel.

TIP: Changes made to a Fuse in a text editor do not immediately propagate to other instances of that Fuse in the composition. Reopening a composition updates all Fuses in the composition based on the current saved version. Alternatively, you can click the Reload button in the Inspector to update the selected node without closing and reopening the composition.